Lochan Reddy

lochanreddy@berkeley.edu | linkedin.com/in/lochannreddy | github.com/lotionr

Education

UNIVERSITY OF CALIFORNIA, BERKELEY Expected Graduation: May 2028

B.A. Computer Science and Data Science *GPA: 3.75*

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Math & Probability Theory, Structure and Interpretation of Computer Programs, Linear Algebra, Foundations of Data Science, Machine Learning, Computer Architecture, Optimization (planned)

Experience

Amazon May 2026 – Aug 2026

Software Development Engineer Intern

- Architected **Pryon 2**, a modular **C++** engine for **real-time inference** (cloud-side wakeword verification on Alexa devices), replacing a legacy system to cut latency and memory overhead across multiple teams; built a custom **JNI** binding layer integrating the engine into Java service infrastructure via a purpose-built API.
- Benchmarked Pryon 2 against the legacy engine, achieving a **64% reduction in latency**; built a **bit-exact validation framework** across **5,844 matched detections**, confirming production readiness for deployment across Alexa device families.
- Independently built a second **C++** engine for **cloud-side audio fingerprint detection**, preventing false wakes from media that would stream user audio without consent — a safety-critical privacy concern across millions of Alexa devices; designed the full JNI and C++ API layer, **adopted by 3+ internal teams**.

Outsampler — Financial AI Research Dec 2025 – May 2026

Machine Learning Research Intern

- Fine-tuned a 3B-parameter multimodal SLM (**Qwen2.5-VL**) with **LoRA** to distill reasoning from GPT-4o, reaching **94%+ citation correctness** for safety-critical model interpretability in high-stakes financial contexts.
- Designed hybrid objective functions (**LIT-C**) aligning spatio-temporal embeddings with analyst reports; improved OOD detection across 7 anomaly families by **15%**; benchmarked via ADBench protocol across **26 real-world datasets**.

Filmcrowd.ai — Media & Entertainment Jan 2026 – May 2026

ML Engineering Intern (Generative AI / Computer Vision)

- Built an automated product placement pipeline using **SAM 2** for instance segmentation and **CLIP** for zero-shot asset selection; engineered a spatio-temporal warping engine via Farneback optical flow and Delaunay triangulation to maintain 3D geometric consistency, deployed as a **Flask REST service**.

Projects

Nanoserve — LLM Inference Engine & Serving System | *C++20, NEON SIMD, Metal, pybind11, Python* Aug 2026

- Built a **zero-dependency C++ inference engine** for Qwen2.5-0.5B from scratch (github.com/lotionr/nanoserve): greedy decoding **bit-identical to HuggingFace**, paged KV cache (**21x lower memory**), OpenAI-compatible SSE serving with **continuous batching (2.85x throughput)**.
- Optimized CPU decode **14x** (307 → 21.9 ms/token) via **NEON SIMD**, threading, and allocation-free decode, then **2x further with INT8 quantization** (integer-sdot GEMV kernels, verified against fp32 goldens) — within **8% of llama.cpp** on a reproducible benchmark harness.
- Implemented a **Metal GPU backend** with zero-copy unified-memory weight residency; measured that dispatch latency dominates GEMV at 0.5B scale — a worked **memory-bandwidth vs. compute-bound** analysis, with CI on Ubuntu/macOS.

Tharros — Real-Time AI Interview Coach | *Python, FastAPI, Overshoot API, ElevenLabs, Mux, Next.js* May 2026

- Built a real-time interview coaching system solo in 4 hours at an a16z hackathon that streams live webcam frames to a vision-language model (**VLM**) detecting behavioral signals (eye contact, posture, nervous tells) at **under 200ms latency**.
- Ran a parallel audio transcription pipeline tracking filler words mid-speech; delivered live voice coaching via ElevenLabs with **sub-500ms end-to-end latency**; recorded sessions via Mux with timestamped playback — **won the multimodal award**.

Dark Pattern Auditor [DEMO] | *Python, FastAPI, Playwright, Perplexity sonar-reasoning-pro, Next.js, Tailwind CSS* Apr 2026

- Built a full-stack agentic legal audit tool that autonomously navigates signup, cancellation, and cookie-consent flows using **Playwright** browser automation, capturing screenshots and page text at each step.
- Classified patterns against **live FTC and EU statute text** via Perplexity sonar-reasoning-pro with real-time `fetch_url` tool calls; streamed crawler progress via **FastAPI SSE** to a Next.js report frontend — demoed at a Perplexity hackathon.

Technical Skills

Languages: Python, C, C++, Swift, Java, JavaScript, SQL, HTML/CSS, Bash, R, Verilog

Frameworks/Libraries: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, React, Flask, SwiftUI

AI & ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models (LLMs), **Real-Time Inference**, **Model Quantization (INT8)**, **Model Distillation**, Fine-tuning (LoRA), Diffusion Models, GNNs, Agentic AI Systems

Tools/Platforms: Git, Docker, AWS SageMaker, Linux (Ubuntu), Bash/Unix CLI, Xcode, GitHub Actions (CI)

Systems: Low-Latency Systems, **SIMD (NEON)**, **Performance Benchmarking**, Memory/Resource Management, Multithreading, Concurrent Systems, REST APIs, Distributed Systems, Unit Testing